

MATH 345 Skeleton Notes

Unit 1: Vectors, matrices, and linear maps

Section 1.6: Exercises

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Exercise tags. The tags below identify the primary outcomes and competency types for each exercise group. Individual problems may also involve earlier outcomes.

Matrices

Primary outcomes. U1-LO3. *Competencies.* P.

- Nicholson 2.1.3(f)
- Nicholson 2.1.8(b)
- Nicholson 2.1.15(b)

Matrix-vector multiplication

Primary outcomes. U1-LO4. *Competencies.* P+R.

- Nicholson 2.2.2(b)
- Nicholson 2.2.3(b)
- Nicholson 2.2.11(d)

Linear and affine maps

Primary outcomes. U1-LO5, U1-LO7. *Competencies.* P+C+R.

- Nicholson 2.6.2(b)
- Nicholson 2.6.13(b)

Matrix multiplication

Primary outcomes. U1-LO6. *Competencies.* P+R.

- Nicholson 2.3.1(b)
- Nicholson 2.3.3(b)
- Nicholson 2.3.5(b)
- Nicholson 2.3.16(b)

Additional applied and computational problems

These exercises connect the main Unit 1 ideas: vectors as data, cosine similarity, weighted averages, matrix-vector products, geometric matrix actions, composition, affine maps, and code interpretation. Solutions are collected in Appendix Unit 1 applied, geometric, and computational interpretation.

Exercise: Word-count vectors and cosine similarity

Use the dictionary `linear`, `matrix`, `data`. Let

$$\mathbf{q} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{D}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{D}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{D}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

1. Compute the cosine similarity between $\mathbf{q}$ and each document vector.

2. Rank the documents by cosine similarity.
3. Explain why $\mathbf{D}_1$ has cosine similarity 1 with $\mathbf{q}$.
4. Which document is orthogonal to $\mathbf{q}$?
5. What does that orthogonality mean in terms of the dictionary words?
6. Compute the Euclidean distance from $\mathbf{q}$ to each $\mathbf{D}_i$. Does the distance ranking match the cosine ranking?

Tags. [U1-L01, U1-L02 | C+R | Core]

Exercise: Weighted average of value vectors

Let

$$\mathbf{v}_1 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 10 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 10 \\ 10 \end{bmatrix},$$

and let

$$\alpha = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{2} \end{bmatrix}.$$

1. Compute $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3$.
2. Explain why this is a weighted average.
3. Which value vector receives the largest weight?

Tags. [U1-L08 | P+C | Core]

Exercise: Attention bridge: hand scores versus $K\mathbf{q}$

Use the token attention activity with tokens “small”, “red”, “bird”. Let

$$\mathbf{q} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$\mathbf{k}_{\text{small}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{k}_{\text{red}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{k}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the three dot products

$$\mathbf{q} \cdot \mathbf{k}_{\text{small}}, \quad \mathbf{q} \cdot \mathbf{k}_{\text{red}}, \quad \mathbf{q} \cdot \mathbf{k}_{\text{bird}}.$$

2. Put the key vectors as rows of the matrix

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Compute $K\mathbf{q}$.

3. Explain why the entries of $K\mathbf{q}$ are the same scores from part (a).
4. Normalize the scores by dividing by their sum. What is the weight vector α ?

Tags. [U1-L01, U1-L04, U1-L08 | C+R | Core]

Exercise: Rows measure; columns contribute

Let

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

1. Compute $A\mathbf{x}$ using row dot products.
2. Compute $A\mathbf{x}$ as a linear combination of the columns of A .

3. In this example, what does the first row of A measure?
4. In this example, what do the coordinates $2, -3, 4$ tell you in the column-combination view?

Tags. [U1-L04 | P+R | Core]

Exercise: The unit-square visualization

Let

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}.$$

1. Compute $A\mathbf{e}_1$ and $A\mathbf{e}_2$.
2. Compute $A \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
3. Apply A to the four corners of the unit square:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

4. Sketch the image of the unit square.
5. Which two transformed edges leaving the origin are the columns of A ?

Tags. [U1-L05, U1-L08 | C+R | Core]

Exercise: Transformation matching

Match each matrix with the correct geometric description.

$$\begin{aligned} A_1 &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, & A_2 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, & A_3 &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \\ A_4 &= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, & A_5 &= \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, & A_6 &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \end{aligned}$$

1. Reflection across the y -axis.
2. Projection onto the x -axis.
3. Horizontal stretch.
4. Horizontal shear.
5. 90° counterclockwise rotation.
6. Reflection across the line $y = x$.

For each match, briefly justify your answer by describing what happens to $\mathbf{e}_1$ and $\mathbf{e}_2$.

Tags. [U1-L05 | C+R | Core]

Exercise: Composition order: RS versus SR

Let

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

1. Compute RS .
2. Compute SR .
3. Are RS and SR equal?
4. Which product represents “apply S first, then apply R ”?
5. Explain geometrically why the two orders produce different results.

Tags. [U1-L06 | P+R | Core]

Exercise: Affine but not linear

Consider the transformation

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 1 \\ y \end{bmatrix}.$$

1. Compute $T(0, 0)$.
2. Explain why T cannot be written in the form $T(\mathbf{x}) = A\mathbf{x}$ for any matrix A .
3. Write T in the affine form

$$T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}.$$

4. Is T linear, affine but not linear, or neither?

Tags. [U1-L07 | C+R | Core]

Exercise: Code interpretation: sorting similarity scores

Consider the code:

```
import numpy as np

doc_names = np.array(["D1", "D2", "D3"])
scores = np.array([1.0, 0.5, 0.0])
ranking = doc_names[np.argsort(scores)::-1]
ranking
```

1. What is the value of `ranking`?
2. What does `np.argsort(scores)` return?
3. What does `::-1` do?
4. What is being ranked?
5. Is this ranking based on direction, distance, or alphabetical order?

Tags. [U1-L02, U1-L08 | R+T | Core]

Exercise: Transformer shape check

Suppose a sequence of token vectors is stored in a matrix

$$X \in \mathbb{R}^{6 \times 5}.$$

Let

$$W_Q, W_K, W_V \in \mathbb{R}^{5 \times 2},$$

and define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

A later weighting step produces an attention-weight matrix A_{att} , and the attention output is

$$H = A_{\text{att}}V.$$

1. What are the shapes of Q , K , and V ?
2. What is the shape of K^T ?
3. What is the shape of $S = QK^T$?
4. What is the shape of A_{att} ?
5. What is the shape of H ?
6. What does the (i, j) -entry of S compute?
7. Which product forms the weighted averages of value vectors?
8. In this convention, are token vectors stored as rows or columns of X ?

Tags. [U1-L03, U1-L06, U1-L08 | R+T | Core]